

Trainings ▾

Specifications

Marine Applications

Starting Motor Circuit:

Maximum resistance:

12-VDC	0.00075 ohm
24-VDC	0.002 ohm

Starting Motor Circuit:

Maximum Voltage Drop:

12-VDC	1.95-VDC
24-VDC	1.7-VDC

Marine Applications

Unaided Cold Start:

Maximum Ambient Temperature	-7°C [19°F]
Minimum Cranking Speed Required	120 rpm

Industrial Applications

Starting Motor Circuit:

Maximum resistance:

12-VDC	0.00075 ohm
24-VDC	0.002 ohm

Starting Motor Circuit:

Maximum Voltage Drop:

12-VDC	1.95-VDC
24-VDC	1.7-VDC

Industrial Applications

Unaided Cold Start:

Maximum Ambient Temperature	-12°C [10°F]
-----------------------------	--------------

Minimum Cranking Speed Required	120 rpm			
Generator Drive Applications				
Minimum Ambient Temperature for Cold Start With Coolant Heated to Rated Speed	-14°C [7°F]			
Minimum Ambient Temperature for Unaided Cold Start to Low Idle Speed	-4°C [25°F]			
Minimum Ambient Temperature for NFPA 110 Cold Start (90°F minimum coolant temperature)	0°C [32°F]			
Minimum recommended battery capacity:				
System Voltage	Ambient Temperature			
	-18°C [0°F]		0°C [32°F]	
	Cold Cranking Amperes	ReserveCapacity ¹ Minutes	Cold Cranking Amperes	ReserveCapacity ¹ Minutes
12-VDC	1800	640	1280	480
24-VDC ²	900	320	640	240
1. The number of plates within a given battery size determines reserve capacity. Reserve capacity determines the length of time that sustained cranking can occur. 2. Cold Cranking Amperes rating are based on two 12-VDC batteries in series.				
For automotive applications a minimum of 9-VDC at the power connector is required to power up to the ECM. For Industrial, Power Generation, and Marine Applications a minimum of 6.5-VDC at the power connector is required to power up the ECM.				

Batteries (Specific Gravity)	
Specific Gravity at 27°C [81°F]	State of Charge
1.260 to 1.280	100%
1.230 to 1.250	75%
1.200 to 1.220	50%
1.170 to 1.190	25%
1.110 to 1.130	Discharged
Maximum Resistance of Starting Motor Circuit	
12-VDC Starting Motor (Ohm) 0.00001	
24-VDC Starting Motor (Ohm) 0.00020	

Cable resistances can be obtained in the accompanying Battery Cable Resistance Chart. If the frame is in ground circuit the frame length must be considered cable of the same size as that used in the balance of the system.

